

Prerequisite Gate — Async Python (before Document 09)

This is for self-study and live mentoring. Full plan: [CURRICULUM.md](#)

Not a numbered document — a short, focused checkpoint. LangGraph's way of running things (streaming, running steps at the same time, `ainvoke / astream`) assumes you're comfortable with `async / await`. This is the first place in the curriculum where you actually need it, so it's taught here instead of earlier, where it would have no real use case yet.

Prerequisites

[08_rag](#)

How to Read & Practice This Gate

- **What:** running several things at once in Python — `async / await`, the event loop, `asyncio.gather`.
- **Why:** LangGraph, and any real production API, depend on this. Skip it, and Doc09 will feel like magic instead of something you understand.
- **When:** any time you're making several separate calls (API calls, database queries) that don't depend on each other's answers.
- **How to practice:** 1. Read the Real Python walkthrough in full — this idea really does need the longer explanation, not just the short official-docs version. 2. Do the **Basic** exercise closed-book. 3. Do the **Intermediate/Real-world** timing comparison — the actual numbers, not just the theory, are what make this click. 4. Do the **Failure** exercise on purpose — breaking things at running-together-ness on purpose teaches this faster than just reading about it. 5. Use **Hint 1** → **Hint 2** only after really trying. Ask for the full solution only if you say "**Show me the solution.**" 6. Before moving on, explain out loud why `async` helps for waiting-on-something work, and does nothing for heavy-computing work. If you can't, you're not done.

Jump to: [Core Concepts](#) · [Practice Exercises](#)

The Story — what this document is actually building

Picture this: Doc09's LangGraph agent needs to call three tools to answer one question — maybe a search, a lookup, and a calculation. Nothing about those three calls depends on the other two's answers. Run them one after another with plain, blocking Python, and you wait for all three, back to back — a 2-second call, another 2-second call, another 2-second call, six seconds total, even though the computer was just sitting idle waiting for a network reply the whole time.

That idle waiting time is the whole story of this gate. `async / await` doesn't make Python faster at computing — it lets Python use the time it would otherwise spend just waiting on a network reply to go start another task instead. One thread, one event loop, switching between tasks only at the moments one of them is waiting on something. `asyncio.gather` is the tool that actually captures that benefit: instead of `await a(); await b(); await c()` (one after another, six seconds), you write `await asyncio.gather(a(), b(), c())` (all three started at once, roughly two seconds — however long the slowest one takes alone).

This has no Build Task of its own — it's a checkpoint, not a project. The reason it exists as its own short document, right here and not earlier, is that Doc09's LangGraph relies on this directly: streaming, running nodes in parallel, and the async client patterns you'll use from here through the rest of the curriculum all assume you're already comfortable with `async / await`. Get this solid now, and Doc09 will read like ordinary code. Skip it, and Doc09 will feel like magic you can't debug.

Core Concepts (read this first — everything you need is here)

The event loop: one thread, many waiting tasks

Python's `asyncio` runs a single-threaded **event loop** that juggles several tasks by switching between them whenever one is *waiting* on something (a network reply, a timer) instead of actively doing work. A coroutine (an `async def` function) is a function that can pause at an `await` point, hand control back to the loop, which goes and makes progress on something else, then comes back to this function once whatever it was waiting for is ready. **Why this is different from using threads:** no separate operating-system threads, and none of the locking complexity that comes with them — running things "at the same time" comes from cooperatively switching at `await` points, all on one thread. **How `await` really behaves:** calling an async function *without* `await` doesn't actually run it — it just creates a coroutine object sitting quietly in memory. `await` is what actually schedules and runs it, which is exactly why "I called the function, but nothing happened" is such a common mistake.

Why async helps waiting-on-something work, and does nothing for heavy-computing work

While your code is waiting for a network reply, the CPU isn't doing anything — it's just sitting idle, waiting. Async lets the event loop use that idle time to make progress on a *different* task instead of just sitting there. **Why this doesn't help heavy-computing work** (lots of math, not waiting on anything outside): there's no idle time to use — the CPU is already fully busy. Switching to another

task wouldn't speed anything up — it would just interleave two computations on the same single thread, taking the same total time (or a bit longer, from the switching itself). **When to actually use `async`:** any code making several separate network calls (several LLM calls, several tool calls, several database queries) that don't depend on each other's answers.

`asyncio.gather` : running things at the same time instead of one after another

`await a(); await b(); await c()` runs three tasks one after another — the total time is roughly the sum of all three. `await asyncio.gather(a(), b(), c())` starts all three *at the same time* and waits for all of them to finish — the total time is roughly however long the *slowest* one takes alone, not the sum of all three. **Why this matters, in real terms, for LLM apps:** if you need to call three separate tools, or check three separate sources of data, running them at the same time instead of one after another can turn a 6-second wait into a 2-second wait — for basically no extra cost, just writing `gather` instead of three separate `await` lines.

Sync vs. async client: same tool, different way of running

The OpenAI library ships two client types — `OpenAI` (runs one thing at a time, waits) and `AsyncOpenAI` (runs things at the same time) — with almost the same method names. The async version just needs `await` in front of each call, and has to run inside an `async def` function under an event loop. **Why both exist:** a simple terminal script (Project 1) doesn't need the extra complexity of async — doing one thing at a time is fine. A production API service handling many requests at once (Doc12) benefits a lot from async, since it can work on other requests while waiting on a slow LLM call, instead of freezing everything.

Common mistakes, and why they're sneaky

Forgetting `await` — as above, the coroutine gets created but never actually runs. This often fails *silently* (no error, just nothing happens), instead of crashing, which makes it unusually hard to notice.

Blocking the event loop — calling a normal, waiting-style function (like the sync `OpenAI` client, or `time.sleep()`) from inside an `async def` function freezes the **whole** event loop for that time — killing the "at the same time" behavior you thought you had, silently — no error, everything just runs one-at-a-time again, for reasons that aren't obvious from just reading the code.

Go Deeper (Optional)

You don't need any of these to understand the Core Concepts above — use them if you want a second explanation or more detail.

- [Python docs — asyncio](#) — the official reference; skim for the shape of it, don't try to memorize every function.
- [Real Python — Async IO in Python: A Complete Walkthrough](#) — a longer, clearer explanation.

Practice Exercises

Setup for this document's practice code: work inside `08b_async_prereq/` (same venv as before — if it's not active, `source .venv/bin/activate`). New package for this document: `pip install openai`.

How to run each exercise: group your practice code by topic, not by difficulty level. If two exercises below are really about the same thing, save them together in ONE script named after that topic — for example, if two exercises are both about `.env` config, save both in one file like `env_config_practice.py`, with each level's version as its own clearly labeled section inside it. Run each topic's file directly, for example: `python env_config_practice.py`.

For this document: - Basic (`first_coroutine`) is its own topic — save it as `coroutine_basics_practice.py` . - Intermediate (`gather_speed`) and Edge cases (`gather_waits_for_slowest`) are both about how `asyncio.gather` behaves — save them together as `gather_practice.py`, with each level as its own section. - Real-world (`async_client_conversion`) is its own topic — save it as `async_client_conversion_practice.py` . - Failure (`blocking_event_loop`) is its own topic — save it as `blocking_event_loop_practice.py` .

Why group by topic instead of by level: if you save each exercise by difficulty level instead, the different versions of the same idea end up scattered across separate files, and you can never see how one topic grows from simple to harder in one place. Grouping by topic keeps that growth visible — open one file, and you see the whole journey for that one thing, from basic to advanced, side by side.

Jump to an exercise: [Basic](#) · [Intermediate](#) · [Real-world](#) · [Edge cases](#) · [Failure](#)

Basic — your first coroutine

- **What:** an `async def` function that awaits `asyncio.sleep(1)` and returns a value, called with `asyncio.run()` .
- **Why:** you need to see, once, that `async def` alone doesn't run anything — `await` and `asyncio.run()` are what actually make it go.
- **When you'll hit this for real:** the first time you write any async code at all, right here.
- **How to code it:** `async def wait_and_return(): await asyncio.sleep(1); return "done"` , then `print(asyncio.run(wait_and_return()))` .
- **Stuck?** [Hint 1](#) · [Hint 2](#) · [Show me the solution](#)

Intermediate — feel the speed difference

- **What:** run 3 API calls one after another with `await` , timed. Then run the same 3 with `asyncio.gather` , timed. Compare.

- **Why:** the numbers, not the theory, are what actually make "running things at the same time" click — this is the exercise that matters most in this whole document.
- **When you'll hit this for real:** any agent making 2+ independent tool calls — this exact pattern is what LangGraph's parallel node execution relies on.
- **How to code it:** write 3 `async def` functions that each `await asyncio.sleep(2)`. Time `await f1(); await f2(); await f3()` with `time.perf_counter()`, then time `await asyncio.gather(f1(), f2(), f3())` — compare the two durations directly.
- **Stuck?** [Hint 1](#) · [Hint 2](#) · [Show me the solution](#)

Real-world — convert a real function to async

- **What:** convert one function from Project 1 (Doc04) to use `AsyncOpenAI` instead of the normal client.
- **Why:** this is the exact conversion Doc12's FastAPI service needs for every route that calls the model — better to do it once here, deliberately, than for the first time under pressure later.
- **When you'll hit this for real:** wrapping any agent behind a real API service (Project 5).
- **How to code it:** replace `from openai import OpenAI` with `from openai import AsyncOpenAI`, make the function `async def`, and `await client.chat.completions.create(...)` instead of calling it directly.
- **Stuck?** [Hint 1](#) · [Hint 2](#) · [Show me the solution](#)

Edge cases — proving `gather` actually waits

- **What:** call `asyncio.gather` with one coroutine that finishes fast and one that sleeps for 5 seconds — confirm `gather` waits for the slow one, and that the fast one's result was ready long before that.
- **Why:** it's easy to assume `gather` returns as soon as the *first* task finishes — it doesn't, and you need to have watched it wait for the slowest one, on purpose.
- **When you'll hit this for real:** any parallel-agent design (Doc11) where one agent is much slower than the others — the whole group waits for it.
- **How to code it:** one coroutine sleeps 0.1s and returns immediately with a printed timestamp, the other sleeps 5s. Run both through `gather`, and print timestamps before/after to see the fast one finished long before `gather` actually returned.
- **Stuck?** [Hint 1](#) · [Hint 2](#) · [Show me the solution](#)

Failure — accidentally blocking the event loop

- **What:** call a normal, waiting-style function from inside an `async def` function on purpose, without `await` or threading, and watch it kill the "at the same time" behavior you expected.
- **Why:** this is the single most common real async bug, and it fails *silently* — no error, just quietly-worse performance — so you need to have seen it once to recognize it later.

- **When you'll hit this for real:** accidentally using the sync `OpenAI` client (instead of `AsyncOpenAI`) inside an async FastAPI route in Project 5.
- **How to code it:** inside an `async def`, call `time.sleep(3)` (not `asyncio.sleep`) instead of awaiting something, run it alongside another coroutine with `gather`, and time it — confirm the total time is now the *sum*, not the max, proving concurrency broke.
- **Stuck?** [Hint 1](#) · [Hint 2](#) · [Show me the solution](#)

Expected Behavior

- You can explain, without notes, why `await asyncio.gather(a(), b(), c())` is faster than `await a(); await b(); await c()` for waiting-on-something work — and why it would *not* be faster for heavy-computing work.
- You can spot, in a piece of code, a call that's silently freezing the event loop.

Break-It / Debug Preview

- A coroutine that's defined but never awaited (it silently does nothing — a very common real mistake).
- A normal, waiting-style call hidden inside an async function that kills "at the same time" behavior without raising any error.
- Full debugging drill in [14_debugging_lab](#).

Interview Topics Preview

- Event loop basics · `asyncio.gather` vs. one-after-another `await` · sync vs. async client trade-offs · what "blocking the event loop" means and why it's bad.

Move On When

You can correctly explain running-at-the-same-time vs. one-after-another async code, and you've converted at least one real function to async without help. Then move on to [09_langgraph](#).

Stuck? Ask for **Hint 1** through **Hint 2**. Ask for the full solution only if you say "**Show me the solution.**"